SINGAPORE MANAGEMENT UNIVERSITY



OPIM 201 Operations Management

SUSTAINABLE SOURCING

Adapted From …

- Chapter 11 “Responsible Purchasing: Moving from Compliance to Value Creation” by Arjan van Weele and Kristine van Tubergen
- Chapter 20 “Improving Social and Environmental Performance in Global Supply Chains” by Hau L. Lee and Sonali V. Rammohan
- Both chapters are from the edited book Y. Bouchery et al. (eds.), 2017 Sustainable Supply Chains, Springer Series in Supply Chain Management
  - Electronic version available in SMU library

Sustainability

- Brundtland definition: "Sustainable Development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs."
  - Proposed in 1987 at the Brundtland Earth Summit chaired by Gro Harlem Brundtland.
  - Hard to operationalize/implement
    - e.g., require predicting the needs of future generations
  - Typically used in tandem with the ESG (Environmental, Social, and Governance) framework
    - ESG framework refers to a set of criteria used to assess a company's environmental, social, and governance impact. Hence it can be viewed as a measured assessment of sustainability.



The Global Supply Chain

- Emerging economies like China and India have risen as major manufacturing and service centers for the global production.
  - Check the labels of your shirt, watch, pen, or notebooks to see where they are made.
- What is the main reason that many western companies (such as Nike, Adidas, Levi's) outsource?
  - Low-cost
- Though outsourcing benefits corporations and consumers, many problems ensued:
  - Myriad social, environmental, and ethical problems at factories which directly or indirectly supply goods and services (called suppliers) to global corporations (called buyers) .
  - Examples:
    - Excessive factory carbon emissions have climate impacts.
    - 200+ million children (10% of children) were engaged in child labor worldwide
    - O'Rourke (2014) highlights the fact that current levels of global production and consumption are using 50 % more natural resources and services than ecosystems generate

Supplier Compliance

- Could the buyer rely on the countries where the suppliers are located to enforce sustainability laws?

Supplier Audit

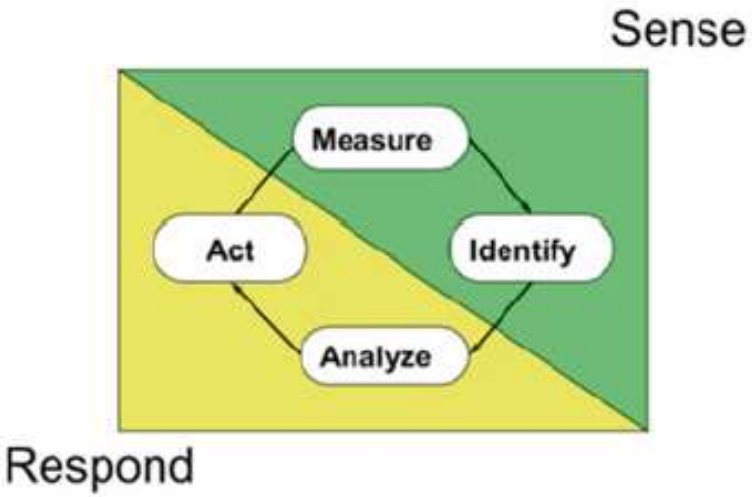
- Is it effective to monitor/audit the suppliers?

Summary

- Key: It is not sufficient to rely solely on self-policing to achieve the desired social and environmental outcome!
- We will examine other approaches for buyers to manage their suppliers.

Sense-Response Framework (Overview)

- A useful guide to address green supply chain management.
  - Haeckel (1992) use it as a system for companies to respond to rapidly changing customer needs.
  - Adopted by Gimenez and Tachizawa (2012)
- Step 1 Sense: A buyer must first gain visibility, or "sense," into issues in the supply chain to understand the present state by measuring and identifying problems
- Step 2 Response: The buyer then "respond" by analyzing the problem and taking action.



Sense-Response Framework (Overview) Continued

- This process can be repeated for continuous improvement.
- This mimics [the familiar] Six Sigma management cycle for eliminating defects and minimizing variability in manufacturing and business processes.

"Sense" in the Sense-Response Framework

- Traceability: ability to trace the points of origin of materials used in a product
  - Need to know exactly where the materials in products come from.
  - Example: Cottons in Levi Strauss & Co was suspected to involve forced child labor.
- Visibility: knowledge of social, environmental, and ethical performance of suppliers
  - Challenge: In a survey by Lee et al. (2012), supply chain executives reported having fairly limited visibility of environmental and social sustainability violations at various levels in the supply network.
- Monitoring: the action of examining supplier performance
  - It alone has limited effectiveness, as mentioned before.

"Response" in the Sense-Response Framework (Overview)

- Lee et al. (2012) summarized six categories (not exhaustive) of key strategies that are commonly used in practice for managing suppliers in terms of sustainability
  - Based on about 1,200 companies

"Response" in the Sense-Response Framework (Strategy 1)

- Strategy 1: Reactions to Violations. Once a violation has occurred, buyers can react with consequences such as root cause analysis and **penalties**.
  - Examples of penalties:
    - Fines
    - Supplier warning
    - Reduced business
    - Contract termination
    - Monetary fines (use of contingencies): a buyer can withhold a portion of payment, which is subject to forfeiture if a supplier violation is found through an audit.

Reaction to Violations

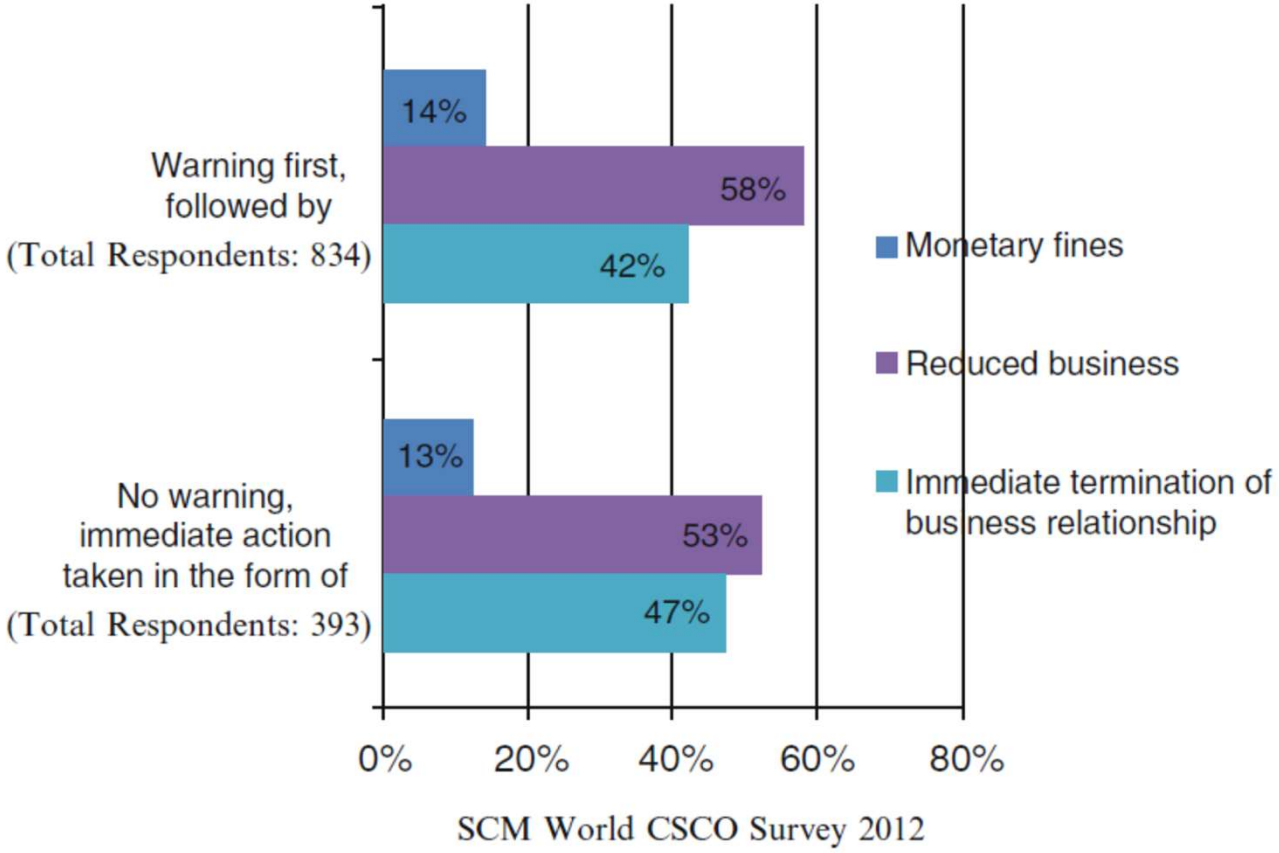


Figure. Reaction to violations (Lee et al, 2012)

"Response" in the Sense-Response Framework (Strategy 1)

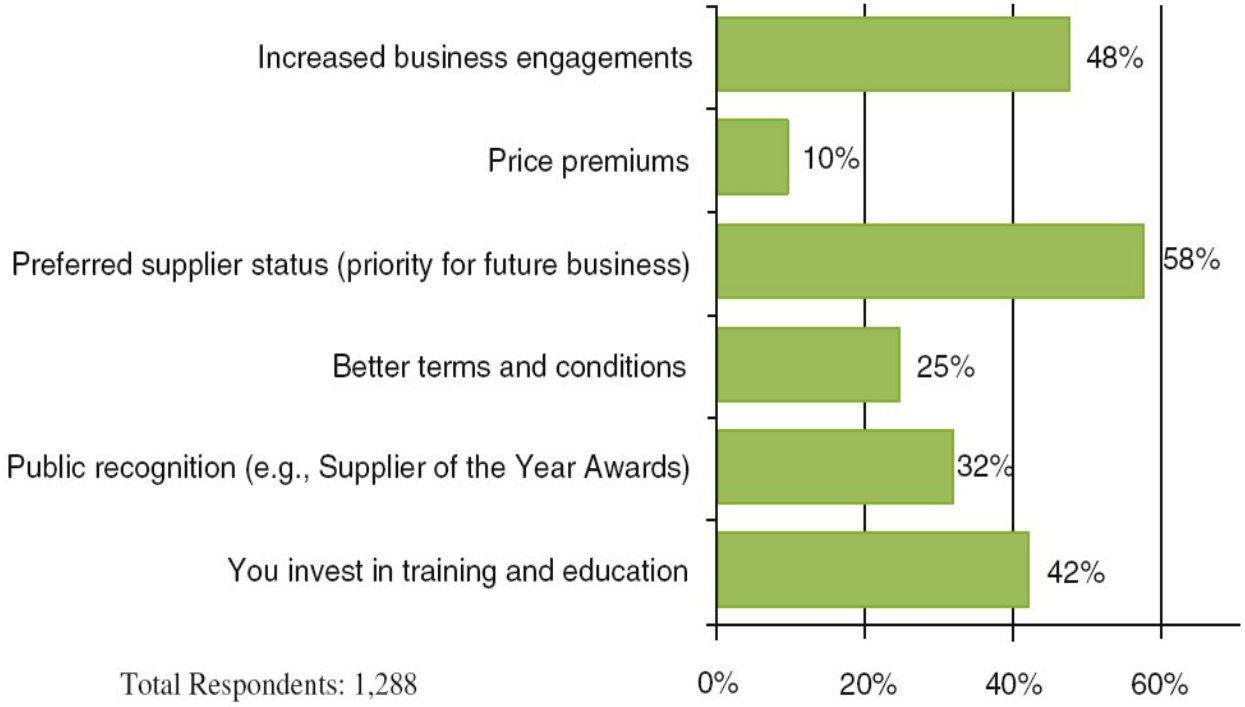
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    - Monetary fines (use of contingencies): a buyer can withhold a portion of payment, which is subject to forfeiture if a supplier violation is found through an audit.
- Lee et al. (2012) find:
  - Monetary penalty were not as common as reducing or terminating business relationships.
  - Many companies have a "zero tolerance" policy for serious issues such as child labor and will terminate business relationships if such issues are detected.
  - About 2/3 of companies surveyed give a warning first before such termination actions.

"Response" in the Sense-Response Framework (Strategy 2)

- Strategy 2: **Incentives** are increasingly used to motivate suppliers to invest in social and environmental improvements.
  - Lee et al (2012) identify six categories of incentives:
    - Supplier training (e.g., management development, lean, six-sigma)
    - Increased business engagements
    - Preferred supplier status
    - Better terms and conditions in the supply contract
    - Public recognition of the supplier (e.g., supplier of the year awards)
    - Price premiums (i.e., buying company pays more for the supplier's product).
- Example:
  - Starbucks' C.A.F.E. Practices: Positive incentives like preferred supplier status as well as price premiums to reward coffee farmers that achieve high sustainability standards.

Frequency of Various Incentives Used in Practice

- Very few companies used price premiums as a reward. Instead, giving suppliers special status, increased businesses, recognition and better terms and conditions are more commonly used.



SCM World CSCO Survey 2012

"Response" in the Sense-Response Framework (Strategy 3)

- Strategy 3: Supplier Collaboration and Capacity Building
  - In the form of education and training, capability enhancement, and the introduction of better production methods for productivity improvement.
- Why can't the buyer leave the suppliers to do that themselves?
  - Suppliers may not have the means and know-how to improve sustainability on their own.
- One of the most prevalent approaches, adopted by companies such as Hewlett-Packard (HP), Li & Fung, Starbucks, McDonald's in India.
- Based on an experiment study of 25 Indian textile plants from 2009 to 2011, the training and support of suppliers significantly increase suppliers' performance.

"Response" In the Sense-Response Framework (Strategy 4)

- Strategy 4: Proactive Product and Process Design
- Examples:
  - Esquel (textile and apparel manufacturer): Pest-resistant cotton seeds
  - McDonald's in India: Potato seeds and growing methods



"Response" in the Sense-Response Framework (Strategy 5)

- Strategy 5: Extended Value Creation and Community Development
- Examples:



Johnson & Johnson



Walmart



Coca-Cola



Unilever



Li & Fung

The Story of Mattel

- In 2007, Mattel had a massive recall of its toys due to the tainting of toys by lead paint.
- The root of the problem was the pigment supplier using pigment-containing lead. The tainted pigments were provided to a paint supplier, whose paint was then supplied to a contract manufacturer who manufactured the toys for Mattel.
- The violation occurred at the third tier of Mattel's supplier network!
- So will auditing the first-tier supplier reveal such problems?
  - No, as the supply network usually has multiple tiers.
- What should Mattel or other companies with many tiers of supply chain do?
  - Cascading Responsible Practices to the Extended Supply Network

"Response" in the Sense-Response Framework (Strategy 6)

- Strategy 6: Cascading Responsible Practices to the Extended Supply Network
- Idea:
  - If each player in the supply chain is responsible for the sustainability of its own supply network, the whole supply chain will be sustainable.
  - Transform suppliers to be as vigilant as the buying firm in managing and assuring sustainability

Examples of Cascading Responsible Practices

- Hewlett-Packard: Engaged intensively with its key suppliers, with the intent of guiding them in being equally focused on sustainability improvements with their own supply networks.
- Intel: Permeate socially and environmentally responsible practices throughout its supply network.
  - The company advocates the concept of supplier ownership of issues, so that suppliers will take a proactive approach in determining their own sustainability strategies and objectives.
  - It has experienced positive results when key suppliers such as Murata and Schneider Electric were able to take on the role of sustainability advocates in managing their own supply network.

Summary of "Response" Strategies in the Sense-Response Framework

- Strategy 1: Reactions to Violations
- Strategy 2: Incentives
- Strategy 3: Supplier Collaboration and Capacity Building
- Strategy 4: Proactive Product and Process Design
- Strategy 5: Shared Value: Extended Value Creation and Community Development
- Strategy 6: Cascading Responsible Practices to the Extended Supply Network

Effectiveness of Strategies 1 and 2

- Survey-based study that examines the effectiveness of Strategy 1 and Strategy 2
  - There are fewer studies on Strategies 3, 4, 5 and 6, and most are case studies.

Research Findings

- Carrot or Sticks? (Incentives or Penalties?)
- Porteous et al. (2015) developed a model of the relationship between the incentives and penalties (issued by buyers to suppliers for social and environmental performance) and improved supplier compliance and buyer-operating costs.
- This model was empirically tested by analyzing opinion-based survey responses from supply chain executives at 334 companies across 17 industries.
- As mentioned in earlier slides, this study did not find a significant relationship between visibility and monitoring efforts (sensing) and reduced violations or reduced operating costs.
- This aligns with research that finds traditional monitoring for violations through supplier audits, when not complemented with supplier collaboration, can be ineffective in addressing persistent supplier violations (Locke et al. 2007).

Research Findings (cont'd)

- Which penalty or incentives have a stronger influence?
  - Examples of penalties:
    - Fines
    - Supplier warning
    - Reduced business
    - Contract termination
    - Monetary fines (use of contingencies): a buyer can withhold a portion of payment, which is subject to forfeiture if a supplier violation is found through an audit.
  - Examples of incentives:
    - Supplier training (e.g., management development, lean, six-sigma)
    - Increased business engagements
    - Preferred supplier status
    - Better terms and conditions in the supply contract
    - Public recognition of the supplier (e.g., supplier of the year awards)
    - Price premiums (i.e., buying company pays more for the supplier's product)

Takeaways

- As rules and regulations in supplier countries are usually lax, (multiple national) companies have to self-regulate.
- Self regulation is not enough, as monitoring alone has limited effect.
- A buyer could consider using a sense-response framework, where the buyer first obtains a sense of the supply chain, and considers response strategies, such as penalty, incentives, and supplier capability building.
- Overall, a buyer has to move away from the traditional cost-driven approach to a value-driven, supplier development approach

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